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$$\int_{-1}^1 \frac{dx}{x^2}$$

Integraal opsplitsen in delen:

$$\int_{-1}^0 \frac{dx}{x^2} + \int_0^1 \frac{dx}{x^2}$$

Linkerintegraal:

$$\begin{aligned} & \lim_{a \rightarrow 0-} \int_{-1}^a \frac{dx}{x^2} \\ &= \lim_{a \rightarrow 0-} \left(-\frac{1}{a} - 1 \right) = -\frac{1}{0-} - 1 = +\infty \end{aligned}$$

Rechterintegraal:

$$\begin{aligned} & \lim_{b \rightarrow 0+} \int_b^1 \frac{dx}{x^2} \\ &= \lim_{b \rightarrow 0+} \left(-1 + \frac{1}{b} \right) = -1 + \frac{1}{0+} = +\infty \end{aligned}$$

Integraal is divergent!

References